

Prepared for

**Reducing Catastrophic Tool Failures
and Excess Inventory Through
Quantitative Optimization**

Executive Summary

Trace3 has established itself as the premier partner for healthcare AI transformation, combining deep NVIDIA expertise with proven healthcare delivery experience. Our portfolio spans 15+ major healthcare implementations, leveraging NVIDIA GPU architecture, validated storage architectures, and clinical-grade AI governance to accelerate diagnostic accuracy, operational efficiency, and patient outcomes. With partnerships across HPE, NetApp, Pure Storage, and exclusive access to NVIDIA's latest technologies including H100, H200, and Blackwell platforms, Trace3 delivers turnkey AI infrastructure that healthcare organizations trust for their most critical workloads.

Why
Trace3 for
Healthcare
AI

15+Healthcare AI
Implementations**DGX +
HPC**Kaiser Permanente
AI Platform**90%**GPU Utilization
Achieved**60K+**Patients Served
(Sutter Continuous
Care)

Strategic NVIDIA & OEM Partnerships

- ▶ **HPE Private Cloud AI (PCAI):** Industry's first turnkey private cloud AI solution, maintained in Trace3 labs for co-developing healthcare patterns and demos
- ▶ **Dell AI Factory:** 25+ year exclusive partnership with co-engineering teams, first-to-market with H100, H200, and Blackwell platforms
- ▶ **NetApp AI Pod:** NVIDIA-validated designs delivering sub-200 μ s latency with DGX H100 integration and air-gapped RAG capabilities for PHI
- ▶ **Pure Storage AIRI/GenAI Pod:** DGX-validated architectures with GPUDirect Storage sustaining 90%+ GPU utilization
- ▶ **Cisco Spectrum-X:** Only networking vendor with silicon integration in NVIDIA's AI networking reference designs
- ▶ **Super Micro:** First-to-market with Blackwell GB200 NVL72 systems achieving 130 TB/s bandwidth

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Major Healthcare Client Implementations

Kaiser Permanente**ThriveAI GPU/HPC Platform**

Architected and implemented a modern AI/ML platform leveraging NVIDIA DGX systems, partnering with HPE for compute, NetApp for storage, and Run:AI for GPU orchestration with Kubernetes. Developed robust testing framework and provided managed engineering to bring use cases onto the platform. Additional work included Snowflake implementation and platform hardening to unify fragmented domains into an AI-ready data foundation.

Sutter Health**Enterprise AI Governance & Continuous Care**

Supported enterprise Continuous Care program integrating Epic and Microsoft to serve approximately 60,000 patients across chronic conditions. Established AI governance with risk-based monitoring, model observability, and intake/registration frameworks aligned to clinical and product teams. CTO recognized Trace3 as a trusted healthcare partner to drive compliance, security, and transformative analytics.

Mary Free Bed Rehabilitation Hospital**Clinical Computer Vision for Early Detection**

Designed and built a computer vision workflow used by clinicians to analyze infant movement videos and accelerate early detection of chronic neurological diseases. Approach included video capture/tagging improvements and ML algorithm development with clinicians, followed by a plan to scale the AI product to affiliates across the rehabilitation network.

Dartmouth Health**Genomics HPC Cluster & AWS Services**

Engaged to move from cloud-based platform to on-prem HPC cluster for genomics. Completed implementation with Slurm-managed, scalable, distributed cluster and continued maintenance/support. Additionally, contracted multi-year AWS/HPC/Genomics services retainer to accelerate genomics pipelines and HPC operations.

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Major Healthcare Client Implementations

Driscoll Children's Hospital

NVIDIA Software

Delivered NVIDIA software to support pediatric AI infrastructure/software needs, enabling advanced clinical applications for children's healthcare.

Medical Liability Insurance Provider

Predictive Analytics & Risk Stratification

Implemented predictive modeling for patient outcome optimization and ML algorithms for risk stratification and operational analytics. Improved SLA response times and enabled advanced healthcare analytics within unique regulatory context.

Sanford Health

AI/ML Strategy & Clinical Enablement

Delivered AI/ML priorities briefing covering data foundation readiness, AI governance/risk controls, solution landscape assessment across neuro care, dermatology AI, surgical workforce efficiency, and Microsoft Copilot readiness. Supporting cloud/network resiliency and security modernization patterns.

HealthEast / M Health Fairview

Clinical Analytics Modernization

Modernized clinical analytics by transforming MNM reporting from quarterly extracts to daily analytics sourced from Epic EHR. Developed HEDIS measure extraction automation and built predictive models to identify underserved populations for community programs including mobile market targeting.

Allina Health

ML-Ready Data Foundations

Optimized data warehouse for ML workloads, significantly reducing ETL processing times. Built Common Data Model supporting the National Patient-Centered Clinical Research Network to enable multi-site AI research and analytics collaboration.

HeartFlow

NVIDIA GPU Enablement

Provided NVIDIA RTX A4000 GPUs for HeartFlow's sites to support AI-enabled medical imaging workloads. Expanded in 2025 to support cardiac imaging analysis and computational fluid dynamics.

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Major Healthcare Client Implementations

BE THE MATCH

Data Quality Health Scorecard + ML

Built Data Quality Health Scorecard providing 360-degree view of constituents, behavioral prediction models for donor engagement, and AI-driven data quality monitoring/anomaly detection to enhance program outcomes and operational reliability.

HealthPartners / University of Minnesota Health

BI to AI Transformation

Migrated legacy BI estates (Crystal Reports/Tableau and SAP Universes) to Power BI with embedded AI capabilities. Delivered best-practice training, enabling modern analytics and AI-augmented reporting across the enterprise.

Advocate Aurora Health

Power BI Enablement & Training

Rolled out Power BI with extensive training across the organization. Strategic implementation revolutionized data analysis and visualization approach, empowering team members with cutting-edge tools and setting new standard for data-driven decision-making.

Ridgeview Medical Center

Cloud Data Analytics Assessment

Comprehensive data assessment evaluating existing data infrastructure, focusing on cloud data analytics capabilities. Strategic recommendations for enhancing data storage, processing, and analytics, ensuring improved efficiency, scalability, and data-driven decision-making.

Healthcare AI Thought Leadership

Trace3 regularly organizes and hosts **AI in Healthcare** sessions with leaders from Providence, Permanente Medical Group, Intermountain, Mayo Clinic, and others, facilitating discussions on clinical AI scaling, governance, and infrastructure patterns. Our team maintains deep relationships with healthcare CIOs, CTOs, and clinical leadership, ensuring our solutions align with real-world operational needs and regulatory requirements.

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NVIDIA-Validated Healthcare Platform Architectures

NVIDIA DGX/HGX + Run:AI + Kubernetes

Pre-validated DGX infrastructure with Run:AI GPU scheduling and Kubernetes orchestration to accelerate training/inference throughput on healthcare use cases (diagnostics, point of care, contact center, pharmacy).

Weka WEKApod

Certified for NVIDIA DGX SuperPOD. AI-native data platform appliance, DGX SuperPOD certified on DGX H100, delivering extreme throughput and scale for clinical imaging, genomics, and video-heavy pipelines.

NetApp AI Pod / FlexPod AI

NVIDIA-Validated Designs with DGX-integrated converged solutions (sub-200µs NFS/RDMA, 100Gbps networking) with Kubernetes-native DataOps, model traceability, and air-gapped RAG options for sensitive PHI.

Pure Storage AIRI / GenAI Pod

NVIDIA-Optimized Storage with DGX-validated architectures featuring GPUDirect Storage and NeMo/NIM software integration that sustain 90%+ GPU utilization and drive significant TCO reductions.

HPE Private Cloud AI (PCAI)

Co-Developed with NVIDIA. Turnkey private cloud AI featuring NVIDIA L40S/H100/GH200 support and Spectrum-X Ethernet integration. Trace3 maintains PCAI in our lab to co-develop healthcare patterns and demos.

Dell AI Factory with NVIDIA

Full-stack AI deployments and data-to-compute architectures hosted in Trace3 labs, aligning to DGX/H100/H200/Blackwell roadmaps and NVIDIA's AI networking reference designs for clinical-grade scale.

Cisco Spectrum-X

Only networking vendor with silicon integration in Spectrum-X, providing optimized AI networking for high-throughput healthcare workloads with end-to-end visibility and performance.

Super Micro Blackwell Systems

First-to-market with Blackwell GB200 NVL72 systems achieving 130 TB/s bandwidth and 96-GPU rack-scale liquid-cooled configurations for maximum performance density.

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Process & Approach

Summary

This engagement applies Operations Research and decision science to optimize tooling reliability and bench-stock strategy in a high-value manufacturing environment. The objective is to reduce financial risk from catastrophic tool failures, free unnecessary capital tied up in inventory, and improve production continuity through quantitative modeling.

The work moves beyond descriptive analytics and dashboards, focusing instead on stochastic reliability modeling, inventory optimization, and decision-ready outputs that can be operationalized within existing systems and processes.

The engagement is structured into three phases:

1. Reliability and Failure Modeling
2. Inventory and Bench-Stock Optimization
3. Operationalization and Decision Enablement

Each phase builds on the prior and produces tangible, reviewable deliverables.

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Process & Approach

Phase 1 — Reliability & Failure Modeling

Goal

Develop a quantitative understanding of tooling failure behavior and risk by modeling tool lifetimes and failure modes as stochastic processes. This phase establishes the probabilistic foundation required for informed inventory and replacement decisions.

Activities

- Review available historical data related to tool usage, failures, maintenance, and replacement
- Perform exploratory analysis to identify variability, outliers, and systemic patterns
- Fit appropriate reliability distributions (e.g., Weibull, lognormal, gamma) to tool lifetimes
- Identify failure modes exhibiting early-life, random, or wear-out behavior
- Quantify failure probabilities over time and under different operating conditions
- Conduct scenario analysis using Monte Carlo simulation to estimate cost exposure from failures

Deliverables

- Reliability model(s) for key tool classes
- Failure probability curves and risk windows
- Quantified cost exposure associated with tool failures
- Technical summary documenting assumptions, methodology, and findings
- Inputs prepared for downstream optimization modeling

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Process & Approach

Phase 2 — Inventory & Bench-Stock Optimization

Goal

Determine optimal bench-stock levels, reorder policies, and stocking strategies that minimize total expected cost while maintaining acceptable risk levels for production continuity.

Activities

- Translate reliability models into inventory decision inputs
- Define cost structures, including:
 - Scrap costs from catastrophic failures
 - Inventory carrying costs
 - Downtime and disruption costs
- Model current (“as-is”) stocking strategy and baseline performance
- Develop optimization models for:
 - Safety stock levels
 - Reorder points
 - Pooling strategies across sites or divisions
- Evaluate tradeoffs between risk tolerance and capital efficiency
- Run scenario comparisons across conservative, moderate, and aggressive strategies

Deliverables

- Optimized bench-stock and reorder recommendations
- Quantified financial comparison of current vs optimized strategies
- Scenario analysis illustrating risk/cost tradeoffs
- Executive-ready summary of recommended inventory policies
- Documentation suitable for audit and review

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Phase 3 — Operationalization & Decision Enablement

Goal

Translate analytical findings into actionable, repeatable decision support that can be integrated into ongoing operations and planning processes.

Activities

- Design decision logic for ongoing use (e.g., thresholds, alerts, replacement triggers)
- Develop lightweight analytical tools or models for planners and stakeholders
- Define data refresh and model update requirements
- Support integration concepts with existing ERP, maintenance, or asset systems
- Conduct knowledge transfer sessions with client stakeholders

Deliverables

- Decision support artifacts (models, dashboards, or planning tools as appropriate)
- Documented operating procedures for ongoing use
- Integration guidance for existing systems
- Final engagement report consolidating findings and recommendations
- Knowledge transfer and walkthrough sessions

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Process & Approach

Roles & Responsibilities

Trace3 Team

Senior Data Solutions Architect

- Role: Technical lead and solution architect
- Responsibilities:
 - Overall analytical approach and methodology
 - Reliability and optimization model design
 - Stakeholder engagement and technical direction
 - Review and validation of deliverables
 - Allocation: 50% (20 hours per week)

Senior Data Engineer

- Role: Primary execution and implementation resource
- Responsibilities:
 - Data ingestion, preparation, and validation
 - Model implementation and simulation execution
 - Scenario analysis and result generation
 - Development of decision support artifacts
 - Allocation: 100% (40 hours per week)

Client Responsibilities

- Provide access to relevant data sources and subject matter experts
- Participate in review sessions and decision validation
- Support integration discussions and operational adoption

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Aggregated Deliverables View

Across all phases, Trace3 will provide:

- Quantitative reliability and risk models
- Optimized inventory and bench-stock policies
- Financial impact and ROI analysis
- Decision-ready recommendations
- Documentation suitable for operational and executive audiences, integration discussions and operational adoption

Estimated Delivery Timeline: 10-12 Weeks

Budgetary Estimate: \$288,000
Subject to detailed SOW